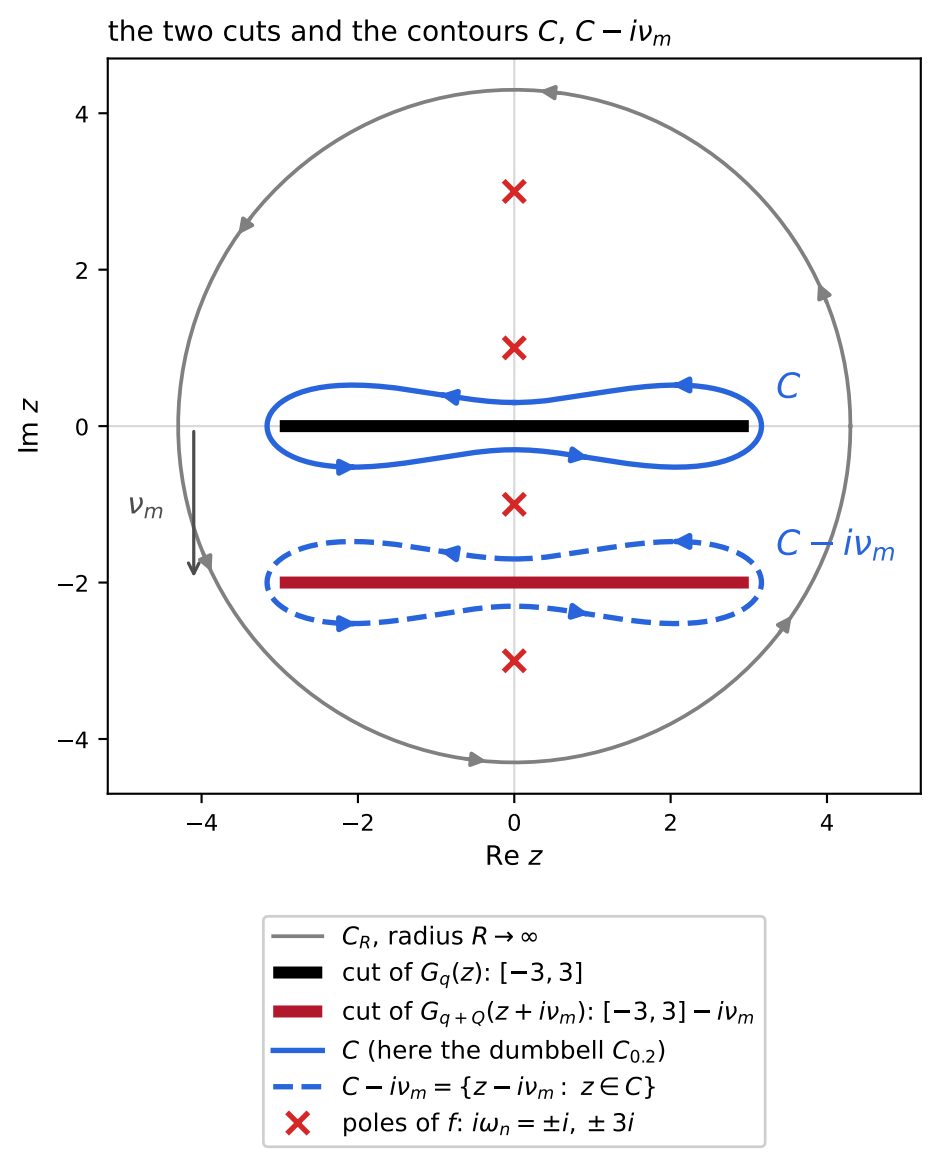
the two cuts and the contours C, $C - i\nu_m$
Im z
Re z
C
$C - i\nu_m$
ν_m
C_R, radius $R \to \infty$
cut of $G_q(z)$: $[-3, 3]$
cut of $G_{q+Q}(z + i\nu_m)$: $[-3, 3] - i\nu_m$
C (here the dumbbell $C_{0.2}$)
$C - i\nu_m = \{z - i\nu_m : \ z \in C\}$
poles of f: $i\omega_n = \pm i, \ \pm 3i$